

Memorial Bridge MVP – Developer System Prompt

Objective

Build the MVP (Minimum Viable Product) for **Memorial Bridge**, a respectful global platform where users can create memorial pages for loved ones and connect those memories to charitable causes.

Architecture Requirements

- **Backend:** Use **Python FastAPI** to develop a RESTful API server.
- **Frontend:** Use **ReactJS** for the web client interface.
- **Mobile:** (*Optional for MVP*) Keep the architecture open for a future **React Native** app, but mobile is not required in this initial POC.
- **Database:** Use **PostgreSQL** (or a similar relational database) to store application data.
- **Image Storage:** Store uploaded images locally during development. Plan to integrate an **S3-compatible storage** for images in the future (for production deployment).
- **Hosting:** Ensure the application can run on **Replit** or a similar low-cost/free deployment platform for demonstration.

Core Functional Scope (MVP)

1. **User Authentication System:**
2. Implement email/password **signup and login** functionality with form validation (e.g. valid email format, password strength).
3. Securely hash user passwords using **bcrypt** before storing in the database.
4. Use **JWT** for session management: issue JWTs upon successful login and require a valid JWT for protected endpoints (e.g. creating a memorial, posting a comment).
5. Include logout functionality (invalidate or expire the JWT on the client side).
6. **Memorial Creation Workflow:**
7. Restrict memorial creation to **authenticated users** only (users must be logged in to create a memorial).
8. Memorial entry fields:
 - **Full Name** of the person being remembered.
 - **Dates** (birth and death dates).
 - **Short Story or Tribute:** A text field for a short biography or tribute (rich text editing is optional; plain text is acceptable for MVP).
 - **Images:** Allow up to **2 images** to be uploaded for each memorial. Include image upload with validation (acceptable file types, size limits) and show a preview of images before submission.
 - **Privacy Setting:** Let the user choose either "public" (visible on public memorial list) or "link-only" (unlisted, only accessible via direct link) for the memorial.

9. After a user submits a memorial, set its status to “**pending approval**”. It should not appear in any public listings until approved by an admin.
10. Create a workflow for an **admin review**: an admin user can view pending memorials and either **approve** (publish it) or **reject** (deny publication) each submission via the admin dashboard.
11. **Memorial Viewing:**
12. Implement a **public dashboard** (homepage or dedicated page) that lists all **approved** memorials that are marked as public. Each listing might show the person's name and maybe a thumbnail or dates, and link to the full memorial page.
13. Implement an **individual memorial page** for each approved memorial. This page should display:
 - The person's **full name** and life **dates**.
 - The **story/tribute** text.
 - The memorial **images** (up to 2, displayed with appropriate size/scaling).
 - If the memorial has associated charities (see Charity Listings below), show a section with those **charitable causes** and provide a static **donation link** or button for each (for MVP, these links can open the charity's website in a new tab).
14. If a memorial is privacy "link-only", do not show it on the public dashboard. Ensure that such a page can still be viewed by anyone with the direct URL (no login required, since it's meant for sharing via link), but it won't be discoverable otherwise.
15. **Comment System:**
16. Enable an authenticated user to **leave comments** on a memorial's page.
17. **Free Users:** Can post **text-only** comments. (The comment input for free users should not allow image uploads.)
18. **Pro Plan Users:** Can post comments that include images, with a limit of **up to 5 images per memorial** per user.
 - Handle the image limit per user/memorial: if a Pro user has uploaded 5 images in comments on a particular memorial, they must delete an existing image comment to free up space for a new one.
 - Provide a UI indication of how many image comments the Pro user has used on that memorial (e.g. "3/5 image comments used").
19. Display comments in **chronological order** (either oldest first or newest first – decide and implement consistently).
20. Implement basic moderation tools for comments:
 - **Auto-Filter:** Automatically filter out or flag profanity and inappropriate language in comments. (This can be a simple keyword blacklist approach for MVP.)
 - **Reporting:** Allow users to **report a comment** for review. For MVP, this can just tag the comment as reported (so an admin can see it in the admin panel and decide to remove it or take action).
21. **Pro Plan Subscriptions (Mocked for MVP):**

22. For the MVP, do **not integrate real payment processing**. Instead, simulate the Pro subscription feature:
- Add a field in the user database (e.g., `is_pro` boolean) to mark Pro users.
 - There will be no user-facing purchase flow; instead, for testing, an admin or developer can manually toggle a user's `is_pro` status in the database.
23. Adjust application logic based on `is_pro`:
- In the frontend, show or enable the image-upload feature in comments only for users marked as Pro. Non-Pro users should see a prompt to upgrade or a disabled upload UI (or simply no option to upload images).
 - In the backend, enforce the rule: if a non-Pro user tries to submit a comment with an image (e.g., via a crafted request), reject it.
24. Provide a way in the admin panel or via a script to mark a regular user as Pro for demonstration purposes (since no payment system exists, this is a manual switch).
25. **Charity Listings (View-Only in MVP):**
26. Implement an admin interface to **manage charities**. Each charity entry includes:
- Charity **Name**.
 - A short **Description** or mission statement.
 - A **URL** (link to the charity's website or donation page).
27. Admin can **create, edit, or delete** charity entries. (For MVP, a simple form in the admin dashboard is sufficient.)
28. Allow users creating or editing a memorial to **associate one or more charities** with that memorial. This could be a multi-select field of available charities (populated from the charity list managed by admin).
29. On an approved memorial's page, if charities are associated, display a **donation section**. For MVP:
- List the associated charity names (each could have a brief description).
 - For each charity, provide a **static donate link/button** that opens the charity's provided URL. (No on-site payment processing is needed in this POC.)
30. (Note: There is no need to track actual donations; this feature is purely informational in the MVP.)

Admin Features

- Implement a simple **Admin Login** mechanism. This could be a separate admin account with a special role or a specific endpoint for admin authentication. (For MVP, even hard-coding one admin user or using an environment variable for an admin password is acceptable, given this is a prototype.)
- Develop an **Admin Dashboard** with the following capabilities:
- **Approve/Reject Memorials:** Show a list of memorial submissions that are in "pending" status. The admin can review each (display basic info and images) and choose to approve or reject.
 - Approving a memorial sets it to "approved" status, making it visible on the site (according to its privacy setting).
 - Rejecting a memorial could either delete it or mark it as rejected; if rejected, optionally provide a way to notify the user or allow them to edit and resubmit (this can be simplified in MVP).

- **Manage Charities:** Provide a section to add new charities and edit or remove existing ones. This should tie into the Charity Listings feature described above.
- **View Users / Pro Status:** Provide a basic view of user accounts. At minimum, show a list of users with an indication of which users are marked as Pro and possibly a way to toggle a user's Pro status. (This helps simulate the subscription feature for demonstration.)
- **Moderate Comments:** (Optional for MVP) If time permits, include a view for reported comments or an interface to remove inappropriate comments. At minimum, ensure admins can see which comments have been flagged by users in the database, even if the UI is rudimentary.

UI/UX Guidelines

- **Overall Design:** Use a **respectful, clean, and accessible design** aesthetic. The tone should be calm and empathetic, given the memorial context.
- **Color Palette:** Opt for calm colors – for instance, light greys, soft blues, and gentle greens. Avoid loud or overly bright colors.
- **Typography:** Use clear, legible fonts. Ensure text, especially the memorial content and names, is easy to read.
- **Layout:** Design a **responsive layout** that works well on desktop and mobile browsers. Key pages include the homepage/public memorial list, memorial detail page, login/signup pages, and the memorial creation form.
- **Memorial Page UI:** Keep the memorial detail pages uncluttered. Focus on the person's **name, photo(s), dates, story**, and the **charity/donation info**. Comments below should be cleanly separated from the main memorial content.
- **Feedback & Alerts:** Provide user feedback for actions:
 - After submitting a memorial, show a confirmation message (e.g., “Your memorial has been submitted and is pending approval.”).
 - After posting a comment, the comment should appear in the list without needing a full page refresh (use React state updates).
 - Show error messages inline for form validation issues (e.g., “Password is too short” or “Image must be JPG/PNG under 5MB”).
- **Navigation:** Include a simple, intuitive navigation:
 - If not logged in: show links to “Home” (public memorials) and “Login/Sign Up”.
 - If logged in: show “Home”, “Create Memorial”, and perhaps a “My Account” or logout button. Admin users should have a link to the Admin Dashboard.
- **Accessibility:** Use semantic HTML elements and ARIA labels where appropriate. Ensure color contrast is sufficient for readability. Users should be able to navigate the site with keyboard and screen readers.
- **Footer:** Include a footer with placeholders for “Privacy Policy”, “Terms of Service”, and “Contact Us” (actual content can be stubbed out in MVP).

Branding & Ethics

- **Tone and Wording:** All user-facing text should be **dignified and empathetic**. For example, instead of a generic success message, use something considerate like “Thank you. Your memorial has been created and is awaiting approval.”.
- **Content Moderation:** Make it clear (perhaps in a tooltip or a note on the memorial creation page) that submitted content is subject to review to maintain a respectful environment.

- **Privacy Respect:** If a user chooses a memorial to be link-only (private), honor that setting strictly. Do not show private memorials or any of their details in public lists or search results. Only those with the direct link (or admins via the dashboard) should be able to view a link-only memorial.
- **No Advertisements or Distracting Elements:** The memorial pages should not contain any ads or promotional content. The focus is solely on the memorial content and the charitable causes.
- **Ethical Data Handling:** Even in a POC, treat user data with care. Use secure connections (HTTPS) in deployment, protect personal data, and ensure that deleting a memorial or comment truly removes the content from public view (you can decide whether to hard-delete or soft-delete in the database for audit purposes, but the content should disappear from the UI when removed).

Notes

- **Donations & Payments:** Do **not implement actual payment processing** or donation transactions in this MVP. All donation links are purely navigational (e.g., linking to an external charity site or a placeholder page).
- **Group Access:** Omit any group-based access or multi-user collaboration features for now. In the MVP, each memorial is managed by its single creator (and admins). Features like inviting family members to manage a memorial can be planned for future versions.
- **Code Quality:** Write clean, modular, and well-commented code. Even though this is a prototype, aim for readability and maintainability, so the project can be built upon in the future.
- **Security & Privacy:** Adhere to basic security best practices from the start (input validation, output encoding, proper authentication checks on each request, etc.). This MVP will serve as a foundation, so it should demonstrate good security principles (e.g., no sensitive data exposure, secure password handling, etc.).

Goal

Produce a **demo-ready application** that stakeholders can interact with to get a feel for the Memorial Bridge concept. The core user flows should be fully functional: - Users can sign up, log in, create a memorial (with text and images), and see their memorial go into a pending state. - Admin can log in to view and approve pending memorials (making them public), manage charities, and oversee user comments. - Visitors (or regular users) can browse public memorials, view individual memorial pages, and post comments (with the system appropriately handling free vs. Pro user capabilities). This MVP should demonstrate the platform's key features and set up a strong foundation for future expansion (such as real payments for subscriptions, enhanced moderation, group memorial management, etc.). The emphasis is on creating a respectful, functional memorial platform connecting memories to charitable causes, with solid architecture and code that can be extended going forward.
